



WHAT IS IN THIS FOLDER

1 Computer Science at Calvin

The three degrees and the cybersecurity certificate. The BCS is accredited by the Computing Accreditation Commission of ABET.

2 Meet the faculty

The six professors who teach the major, what they research, and how to reach any of them directly.

3 Your four years, mapped

A semester-by-semester course plan you can tick off, one sheet per degree.

TAKE THE NEXT STEP



VISIT US

Sit in on a class and see the labs.
calvin.edu/visit



APPLY

calvin.edu/admissions/apply



ASK US ANYTHING

Write to any professor on the faculty sheet.
computing.calvin.edu

QR

Scan for the department site, current student projects, and the courses you would take in your first year. *[Generate the code once the page URL is settled.]*



COMPUTER SCIENCE

DEPARTMENT DISTINCTIVES

Three majors: Bachelor of Computer Science, Computer Science Plus, and Data Science. Add to that minors in computer science, data science, and data analytics, plus a new certificate in cybersecurity. Students work on high-end equipment you'd normally find only at a tier-one research school. Every professor is a committed Christian who brings that faith to bear on their teaching and their code.

BACHELOR OF COMPUTER SCIENCE

Our flagship major is ABET-accredited, meeting the same standards as programs at major research universities. Coursework spans algorithms, artificial intelligence, security, databases, networks, and software engineering. Graduates move into software development, network engineering, and systems work, or on to graduate study at Michigan, MIT, Purdue, Stanford, Texas, and Wisconsin.

COMPUTER SCIENCE PLUS

A full computer science degree paired with a second field such as business, psychology, or language, for students who want to apply computational thinking wherever their interests lie. Our Strategic Partners Council regularly reviews the curriculum so it tracks the technologies employers actually use, and students take paid internships and research alongside faculty.

DATA SCIENCE

Calvin was among the first Christian institutions to offer data science. Students combine computer science, statistics, and mathematics with four to five courses in a second discipline such as biology, geography, business, economics, or political science, then finish with a capstone project built for a professional portfolio.

CYBERSECURITY CERTIFICATE NEW FOR FALL 2026

Eighteen credit hours in cryptography, network defense, and system administration, designed around the NSA's Cyber Defense knowledge units and taught by a professor with three decades in network security. BCS students earn the certificate inside their existing major with no added credit hours. Hands-on labs, projects, and internships throughout. This field is projected to grow 33 percent this decade.





Derek Schuurman

PROFESSOR · CHAIR · NH 296

Focus: Faith and technology, embedded systems and IoT, robotics and computer vision

Email: derek.schuurman@calvin.edu

- PhD, McMaster University
- BAsC and MASc in Electrical Engineering, University of Waterloo
- Writes on the ethics and philosophy of technology



Ken Arnold

ASSOCIATE PROFESSOR · NH 298

Focus: Data science, artificial intelligence and Christian faith

Email: ken.arnold@calvin.edu

- PhD in Computer Science, Harvard University, 2020
- SM, MIT Media Lab; BS in Electrical and Computer Engineering, Cornell
- Language models for writing, and public understanding of what AI can and cannot do



Eric Araújo

ASSOCIATE PROFESSOR · NH 293

Focus: Agent-based modelling, complex networks, computational social science

Email: eric.araujo@calvin.edu

- PhD in Computer Science, Vrije Universiteit Amsterdam, 2018
- MS and BS in Computer Science, Brazil
- Studies how behaviour, health and misinformation spread



Rocky Chang

PROFESSOR · NH 299A

Focus: Network security, internet infrastructure, computational thinking

Email: rocky.chang@calvin.edu

- PhD in Computer and Systems Engineering, Rensselaer Polytechnic Institute
- More than 30 years teaching computer science
- Leads the cybersecurity certificate. Calvin faculty since 2023



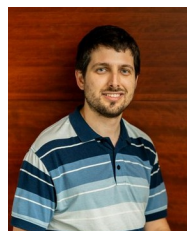
Victor Norman

ASSOCIATE PROFESSOR · NH 297

Focus: Computer science education, K-12 outreach, computing for Christian missions

Email: victor.norman@calvin.edu

- PhD and MS, Purdue University; BS, Calvin College
- 14 years in industry as a principal software engineer before teaching
- Works on applying the virtues to how we write programs



Fernando Santos

ASSISTANT PROFESSOR · NH 291

Focus: Data science, computational modelling, human-computer interaction

Email: fernando.santos@calvin.edu

- PhD in Computational Science, University of São Paulo
- Also works on the philosophy of technology
- Originally from Poços de Caldas, Brazil



Chris Wieringa

SYSTEMS ENGINEER

Focus: The labs, the servers and the systems every course here runs on

Email: chris.wieringa@calvin.edu

- Runs the labs, servers and systems behind every computer science course
- Guides the CS tour on two of this autumn's Discovery Days

ALSO TEACHING WITH US

Rob Bobeldyk Information Technology rob.bobeldyk@calvin.edu

Stacy DeRuiter Mathematics and Statistics stacy.deruiter@calvin.edu

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COMPUTER SCIENCE

ABET ACCREDITED

CHRIST-CENTERED UNDERSTANDING

Technology is never neutral, and we say so out loud. You will study the ethics and philosophy of technology alongside the engineering of it, and ask what a piece of software does to the people who use it. Faith integration with computing is a research area here, not an add-on module.

TAUGHT BY PROFESSORS, ALL SIX OF THEM

Six professors teach the courses in the major, and their offices sit along one corridor in North Hall. You will not be handed off. Their doctorates come from Harvard, Purdue, McMaster, Waterloo, Rensselaer, Amsterdam and São Paulo, and several worked in industry first.

FOUR-YEAR PATH TO GRADUATION

The Bachelor of Computer Science is 124 credit hours. The Calvin Core runs alongside it, so you graduate having read and argued as well as built.

[The number that belongs here is the four-year graduation rate, or the share of students who complete an internship. Engineering leads with both. We do not have ours yet.]

v1.4 for review

ACADEMIC OPTIONS

DEGREES

Bachelor of Computer Science (BCS)
Our most rigorous programme. ABET accredited.

BS in Computer Science Plus
Computer science with an applied field of your choice.

BS in Data Science
Computer science and statistics, paired with a subject area.

MINORS

Data Science
Data Analytics

CERTIFICATE

Cybersecurity
Six courses, 18 credit hours, taken alongside any of the degrees.

PAIR WITH

Biology · Business · Economics · Geography
· Political science · Psychology · Languages



WHAT YOU DO HERE

INTERNSHIPS THAT LEAD SOMEWHERE

Students work a paid internship at a local firm or a national one, and many turn into offers. Recent placements include AMD, Amazon Web Services, General Motors, Corewell Health, Navy Federal Credit Union, Meta, Vervint, Bindplane, EduStaff, Sonset Solutions, Wycliffe Bible Translators and the Christian Classics Ethereal Library.

UNDERGRADUATE RESEARCH

You can join a professor's research rather than wait for graduate school. Current work spans language models and everyday creativity, internet infrastructure and denial-of-service defence, agent-based models of how misinformation and health behaviour spread, computer science education, and embedded systems and robotics.

LIBERAL ARTS AND THE CALVIN CORE

The Core sits alongside the major rather than under it. Graduates leave able to write, argue and place their work in a longer history, which is the difference employers describe when they say someone is easy to work with.

WHERE OUR GRADUATES GO

Alumni are at Google, Amazon, Meta, Steelcase, Corewell Health, Bank of America, the Washington Post, Amway and Pray.com, and in graduate programmes across the country. *[Confirm the graduate school list before print. The version on the website is of unknown vintage.]*



EDUCATION & COURSEWORK

FOUNDATIONS: YEARS 1 AND 2

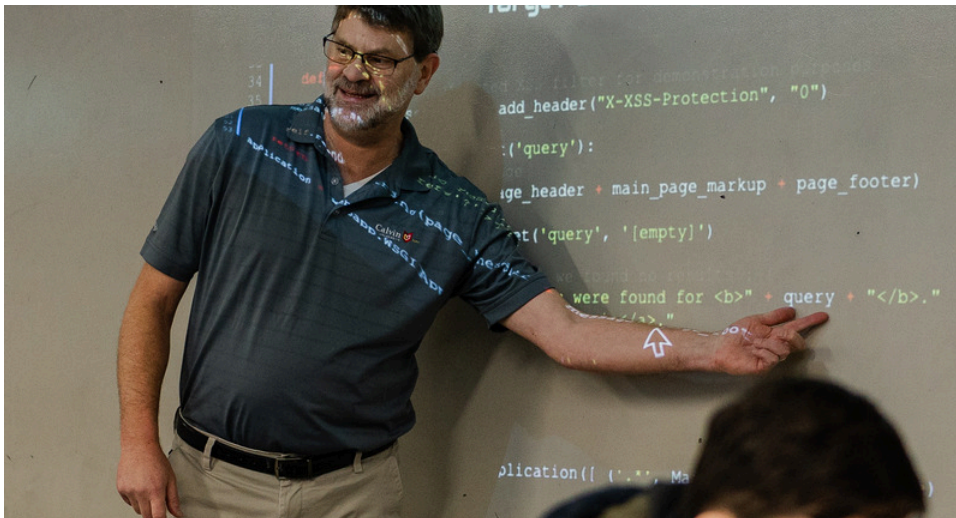
You start writing programs in your first semester, whatever you have done before. The first two years build the craft and the theory underneath it: programming, data structures, the machine below the language, and the mathematics that makes algorithms provable rather than hopeful. You take Core courses alongside, so the questions start early.

[Name the first-year courses here. A family wants to see CS 108 and CS 112 on the page.]



DEPTH: YEARS 3 AND 4

In the upper years you choose the direction: systems and networks, artificial intelligence and data, security, software engineering, or the applied field you paired with. Every student finishes with a senior project built to a real brief, presented and published as a website of its own.



CALVIN HACKATHON

- A weekend, a team, and something that works by Saturday night
- Students also compete off campus. A Calvin pair took first place at the Atomic Object games, ahead of Grand Valley

CYBERSECURITY CERTIFICATE

- Six courses, 18 credit hours, taken alongside any degree
- Built around the Cyber Defense knowledge units published by the NSA and NCAE-C
- Courses run on a rotation rather than all in one semester

SENIOR PROJECT

- A full-year build for a real brief
- Every project ships a public website, and this year a poster too

CLUBS AND COMMUNITY

- Abstraction and Women in Computing
- Game nights, Lego events, scavenger hunts and sticky-note pixel art

The full course list for each degree is on the model program sheets in this folder, one per degree.

MORE THAN THE CODE

At Calvin you write real software from your first semester, and you are asked what it is for. Computer science here is a craft with consequences, so you learn to build well and to think about who your work serves.

PAIR IT WITH ANYTHING

Computer Science Plus lets you take the full computer science degree and add an applied field of your own choosing, planned around another department's minor. Data Science students do the same, pairing with biology, geography, business, economics or political science.

STUDENT ORGANIZATIONS

Student-led clubs put computing to work and build community around it.

- Abstraction
- Women in Computing
- Google Developer Student Club *[confirm current name]*

CONFERENCES AND COMPETITIONS

Students travel with faculty, compete, and meet the field.

- Calvin Hackathon
- Grace Hopper Celebration for Women in Computing
- GrrCon and GR Tech Week
- CCSC and the Create Conference

RESEARCH WITH FACULTY

Undergraduates work alongside professors on real problems: artificial intelligence and language models, network and internet security, agent-based modelling of how behaviour spreads, computer science education, embedded systems and robotics, and the integration of faith with computing.

ADVICE FOR HIGH SCHOOL

[What should a student take before arriving, and do they need to have programmed before? Engineering's sheet answers this in four lines. Ours should too.]

ABET

The Bachelor of Computer Science is accredited by the Computing Accreditation Commission of ABET, www.abet.org

WHAT OUR GRADUATES DO

- Software engineering
- Artificial intelligence and machine learning
- Cybersecurity and computer forensics
- Data science and analytics
- Cloud architecture and infrastructure
- Databases and information systems
- Embedded systems

TAKE THE NEXT STEP

Apply to Calvin



calvin.edu/admissions/apply

To schedule a visit

Call (800) 688-0122

Text (616) 202-3256

admissions@calvin.edu

calvin.edu/visit

Bachelor of Computer Science

Our most rigorous computer science degree, and the one accredited by the Computing Accreditation Commission of ABET.

FRESHMAN

FALL	14 hrs
☐ M 4 CS 108 F,S Introduction to Computing	
☐ M 4 MATH 161 F,S Calculus: Modeling and Computation	
☐ M 0 CS 195 F,S Introductory Computing Seminar · three semesters	
☐ C 2 Community & Commitments CORE 100	
☐ C 4 World Languages I 1 Waived with two years in high school at B or better	

SPRING	15 hrs
☐ M 4 CS 112 F,S Introduction to Data Structures	
☐ M 2 MATH 251 S Discrete Mathematics I	
☐ M 0 CS 195 F,S Introductory Computing Seminar	
☐ C 4 Foundational Writing ENGL 101	
☐ C 4 World Languages I 2	
☐ C 1 Health & Movement 1	

SOPHOMORE

FALL	16 hrs
☐ M 4 CS 212 F Data Structures and Algorithms	
☐ M 4 MATH 255 F Introductory Linear Algebra	
☐ M 4 DATA 202 F Introduction to Machine Learning	
☐ M 0 CS 195 F,S Introductory Computing Seminar	
☐ C 4 Christian Foundations 1	

SPRING	16 hrs
☐ M 4 CS 214 S Programming Language Concepts	
☐ M 2 MATH 252 S Discrete Mathematics II · MATH 251 first	
☐ M 4 STAT 243 S Statistics, or an approved alternative	
☐ C 4 Arts & Rhetoric 1 COMM 101 Oral Rhetoric counts here	
☐ C 2 Christian Foundations 2	

JUNIOR

FALL	16 hrs
☐ M 4 CS 262 F Software Engineering · junior standing	
☐ M 4 ENGR 220 F Introduction to Computer Architecture	
☐ M 0 CS 295 F,S Computing Seminar · three semesters	
☐ C 4 Knowledge & Understanding Tagged · see the Core Options sheet	
☐ E 4 Elective	

SPRING	15 hrs
☐ M 4 CS 232 S Operating Systems and Networking	
☐ M 4 CS elective From the advanced elective list	
☐ M 0 CS 295 F,S Computing Seminar	
☐ C 1 Health & Movement 2	
☐ C 4 Knowledge & Understanding Tagged	
☐ E 2 Elective	

SENIOR

FALL	16 hrs
☐ M 2 CS 396 F Senior Project in Computing	
☐ M 0 CS 295 F,S Computing Seminar	
☐ M 2 CS elective	
☐ M 4 Science elective Laboratory science from the approved list	
☐ C 4 Knowledge & Understanding Tagged	
☐ E 4 Elective	

SPRING	16 hrs
☐ M 2 CS 398 S Senior Project in Computing II	
☐ M 4 CS 384 S Perspectives on Computing · final year	
☐ M 4 CS elective Ten hours in total	
☐ M 2 Science elective Six hours in total	
☐ E 4 Elective	

MAJOR

72

CORE

38

ELECTIVE

14

GRAND TOTAL

124 hours

Computer Science Plus

The full computer science degree, plus twelve hours in an applied field of your own choosing.

FRESHMAN

FALL	14 hrs
☐ M 4 CS 108 F,S Introduction to Computing · or CS 106	
☐ M 4 MATH 161 F,S Calculus: Modeling and Computation	
☐ M 0 CS 195 F,S Introductory Computing Seminar · three semesters	
☐ C 2 Community & Commitments CORE 100	
☐ C 4 World Languages I 1 Waived with two years in high school at B or better	

SPRING	15 hrs
☐ M 4 CS 112 F,S Introduction to Data Structures	
☐ M 2 MATH 251 S Discrete Mathematics I	
☐ M 0 CS 195 F,S Introductory Computing Seminar	
☐ C 4 Foundational Writing ENGL 101	
☐ C 4 World Languages I 2	
☐ C 1 Health & Movement 1	

SOPHOMORE

FALL	16 hrs
☐ M 4 CS 212 F Data Structures and Algorithms	
☐ M 4 DATA 202 F Introduction to Machine Learning	
☐ M 4 Domain study Your applied field · twelve hours in total	
☐ M 0 CS 195 F,S Introductory Computing Seminar	
☐ C 4 Christian Foundations 1	

SPRING	16 hrs
☐ M 4 CS 214 S Programming Language Concepts	
☐ M 2 MATH 252 S Discrete Mathematics II · MATH 251 first	
☐ M 4 STAT 243 S Statistics, or an approved alternative	
☐ C 4 Arts & Rhetoric 1	
☐ C 2 Christian Foundations 2	

JUNIOR

FALL	16 hrs
☐ M 4 CS 262 F Software Engineering · junior standing	
☐ M 4 ENGR 220 F Introduction to Computer Architecture	
☐ M 4 Domain study Your applied field	
☐ M 0 CS 295 F,S Computing Seminar · three semesters	
☐ C 4 Knowledge & Understanding Tagged	

SPRING	15 hrs
☐ M 4 CS 232 S Operating Systems and Networking	
☐ M 4 CS elective Six hours in total	
☐ M 0 CS 295 F,S Computing Seminar	
☐ C 1 Health & Movement 2	
☐ C 4 Knowledge & Understanding Tagged	
☐ E 2 Elective	

SENIOR

FALL	16 hrs
☐ M 2 CS 396 F Senior Project · or CS 383 External Practicum	
☐ M 0 CS 295 F,S Computing Seminar	
☐ M 4 Domain study Your applied field	
☐ C 4 Knowledge & Understanding Tagged	
☐ E 4 Elective	
☐ E 2 Elective	

SPRING	16 hrs
☐ M 2 CS 398 S Senior Project in Computing II	
☐ M 4 CS 384 S Perspectives on Computing · final year	
☐ M 2 CS elective	
☐ E 4 Elective	
☐ E 4 Elective	

MAJOR
70

CORE
38

ELECTIVE
16

GRAND TOTAL
124 hours

Data Science

Computer science and statistics together, applied to a subject area you care about.

FRESHMAN

FALL	14 hrs
☐ M 4 CS 108 F,S Introduction to Computing · or CS 106	
☐ M 4 MATH 161 F,S Calculus · or MATH 163	
☐ C 2 Community & Commitments CORE 100	
☐ C 4 World Languages I 1 Waived with two years in high school at B or better	

SPRING	16 hrs
☐ M 4 CS 112 F,S Introduction to Data Structures	
☐ M 4 STAT 243 S Statistics	
☐ C 4 Foundational Writing ENGL 101	
☐ C 4 World Languages I 2	

SOPHOMORE

FALL	16 hrs
☐ M 4 DATA 202 F Introduction to Machine Learning	
☐ M 4 MATH 255 F Introductory Linear Algebra	
☐ C 4 Knowledge & Understanding Tagged	
☐ C 4 Christian Foundations 1	

SPRING	15 hrs
☐ M 2 CS 375 S Machine Learning	
☐ C 4 Arts & Rhetoric 1	
☐ C 1 Health & Movement 1	
☐ C 2 Christian Foundations 2	
☐ E 4 <i>Elective</i>	
☐ E 2 <i>Elective</i>	

JUNIOR

FALL	16 hrs
☐ M 4 STAT 245 F Applied Data Analysis	
☐ M 4 STAT 343 F Probability and Statistics · after MATH 255	
☐ M 4 CS elective CS 212 recommended · or CS 354, CS 374	
☐ M 4 Business or soft-skills elective Six hours in total	

SPRING	15 hrs
☐ M 2 CS 376 S Advanced Machine Learning	
☐ M 2 DATA 304 S Visualizing Data and Models	
☐ M 4 DATA 385 S Topics in Data Science	
☐ C 1 Health & Movement 2	
☐ C 4 Knowledge & Understanding Tagged	
☐ E 2 <i>Elective</i>	

SENIOR

FALL	16 hrs
☐ M 2 DATA 396 F Senior Project in Data Science	
☐ M 4 Ethics elective PHIL 205 Ethics or BUS 392 Business Ethics	
☐ C 4 Knowledge & Understanding Tagged	
☐ E 4 <i>Elective</i>	
☐ E 2 <i>Elective</i>	

SPRING	16 hrs
☐ M 2 DATA 398 S Senior Project in Data Science II	
☐ M 4 CS 384 S Perspectives on Computing · final year	
☐ M 2 Business or soft-skills elective	
☐ E 4 <i>Elective</i>	
☐ E 4 <i>Elective</i>	

MAJOR

64

CORE

38

ELECTIVE

22

GRAND TOTAL

124 hours

